



# CLASSROOM CONTACT PROGRAMME

(Academic Session : 2022 - 2023)

JEE(Main)

FULL TEST

25-12-2022

JEE(Main+Advanced) : ENTHUSIAST COURSE [PHASE : TOAS, TNAS, TAAS, TEAS, TIAS, I(A) & LIVE-I(A)] (SCORE-I)

Time : 3 Hours

Maximum Marks : 300

## READ THE INSTRUCTIONS CAREFULLY

### Important Instructions :

1. Immediately fill in the form number on this page of the Test Booklet with Blue/Black Ball Point Pen. Use of pencil is strictly prohibited.
2. The candidates should not write their Form Number anywhere else (except in the specified space) on the Test Booklet/Answer Sheet.
3. The Test Booklet consists of **90** questions.
4. There are **three** parts in the question paper 1,2,3 consisting of **Physics, Chemistry and Mathematics** having **30 questions** in each subject and each subject having **Two sections**.
  - (i) Section-I contains 20 multiple choice questions with **only one correct** option.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
  - (ii) Section-II contains 10 **Numerical Value Type** questions. Attempt any 5 questions. First 5 attempted questions will be considered for marking.  
**Marking scheme** : +4 for correct answer, 0 if not attempted and -1 in all other cases.
5. Use **Blue/Black Ball Point Pen only** for writing particulars/markings responses on **Side-1** and **Side-2** of the Answer Sheet. **Use of pencil is strictly prohibited.**
6. No candidate is allowed to carry any textual material, printed or written, bits of papers, mobile phone any electronic device etc, except the Identity Card inside the examination hall/room.
7. Rough work is to be done on the space provided for this purpose in the Test Booklet only.
8. On completion of the test, the candidate must hand over the Answer Sheet to the invigilator on duty in the Room/Hall. **However, the candidate are allowed to take away this Test Booklet with them.**
9. **Do not fold or make any stray marks on the Answer Sheet.**
10. Take  $g = 10 \text{ m/s}^2$  unless otherwise stated.

DO NOT BREAK THESE SEALS WITHOUT BEING INSTRUCTED TO DO SO BY THE INVIGILATOR

Name of the Candidate (in Capitals)\_\_\_\_\_

Form Number : in figures\_\_\_\_\_

: in words\_\_\_\_\_

Centre of Examination (in Capitals) : \_\_\_\_\_

Candidate's Signature : \_\_\_\_\_ Invigilator's Signature : \_\_\_\_\_

**Your Target is to secure Good Rank in JEE(Main) 2023**

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SECTION-I : (Maximum Marks: 80)

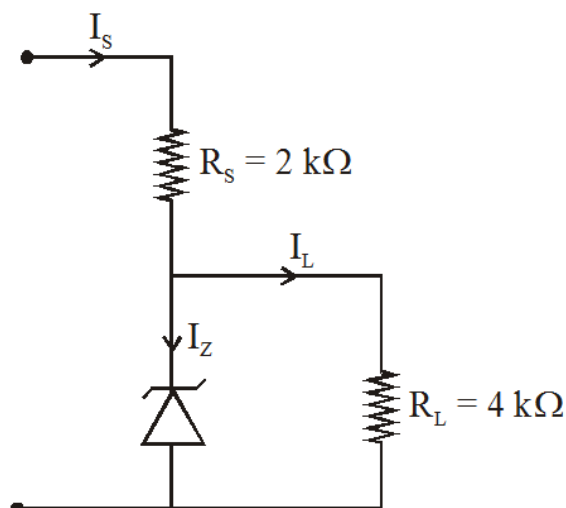
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

**Full Marks** : +4 If correct answer is selected.

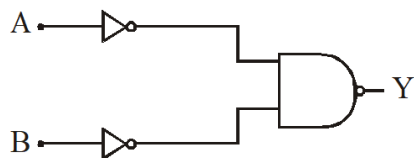
**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. Figure shown a DC voltage regulator circuit, with a Zener diode of breakdown voltage = 6V. If the unregulated input voltage varies between 10 V to 16 V, then what is the maximum Zener current ?



- (A) 2.5 mA (B) 3.5 mA  
(C) 7.5 mA (D) 1.5 mA
2. The logic gate equivalent to the given logic circuit is :-



- (A) OR (B) AND (C) NOR (D) NAND

3. A telephonic communication service is working at carrier frequency of 10 GHz. Only 10% of it is utilized for transmission. How many telephonic channels can be transmitted simultaneously if each channel requires a bandwidth of 5 kHz ?

- (A)  $2 \times 10^4$   
(B)  $2 \times 10^5$   
(C)  $2 \times 10^6$   
(D)  $2 \times 10^3$

4. Unpolarized light of intensity  $I$  passes through an ideal polarizer A. Another identical polarizer B is placed behind A. The intensity of light beyond B is found to be  $\frac{I}{2}$ . Now another identical polarizer C is placed between A and B. The intensity beyond B is now found to be  $\frac{I}{8}$ . The angle between polarizer A and C is :

- (A)  $30^\circ$   
(B)  $45^\circ$   
(C)  $60^\circ$   
(D)  $0^\circ$

5. The angular width of the central maximum in a single slit diffraction pattern is  $60^\circ$ . The width of the slit is  $1 \mu\text{m}$ . The slit is illuminated by monochromatic plane waves. If another slit of same width is made near it, Young's fringes can be observed on a screen placed at a distance 50 cm from the slits. If the observed fringe width is 1 cm, what is slit separation distance? (i.e. distance between the centres of each slit.)

(A)  $50 \mu\text{m}$   
 (B)  $75 \mu\text{m}$   
 (C)  $100 \mu\text{m}$   
 (D)  $25 \mu\text{m}$

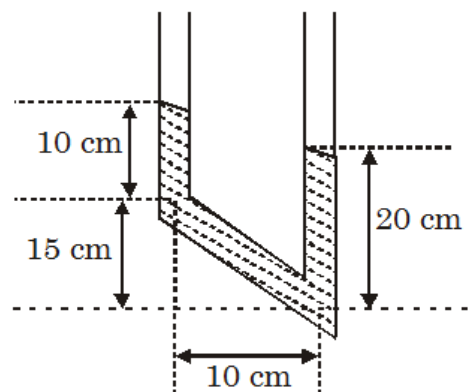
6. Two skaters A and B of masses 50kg and 70kg respectively stand facing each other 6m apart. Then they pull on a rope stretched between them. How far has each moved when they meet?

(A) Both 3m  
 (B) A moves 2.5m and B, 3.5m  
 (C) A moves 3.5m and B, 2.5m  
 (D) A moves 2m and B, 4m

7. One end of a horizontal thick copper wire of length  $2L$  and radius  $2R$  is welded to an end of another horizontal thin copper wire of length  $L$  and radius  $R$ . When the arrangement is stretched by applying equal forces at two ends, the ratio of the elongation in the thin wire to that in the thick wire is :

(A) 0.25  
 (B) 0.50  
 (C) 2.00  
 (D) 4.00

8. A U tube filled with a liquid is accelerating horizontally with an acceleration  $a$ . The acceleration of the tube is (Take :  $g = 10 \text{ m/s}^2$ ) :-

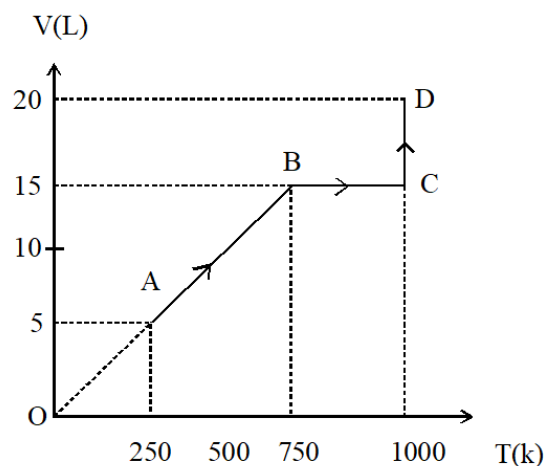


(A) 10  
 (B) 5  
 (C) 6  
 (D) 7

9. A particle is projected with a velocity  $\sqrt{\frac{4gR}{3}}$  vertically upward from the surface of the earth,  $R$  being the radius of the earth &  $g$  being the acceleration due to gravity on the surface of the earth. The velocity of the particle when it is at half the maximum height reached by it is

(A)  $\sqrt{\frac{gR}{2}}$  (B)  $\sqrt{\frac{gR}{3}}$   
(C)  $\sqrt{gR}$  (D)  $\sqrt{\frac{2gR}{3}}$

10. Two moles of helium gas is taken along the path ABCD. Note that line AB extended backwards passes through origin. Total work done by the gas is

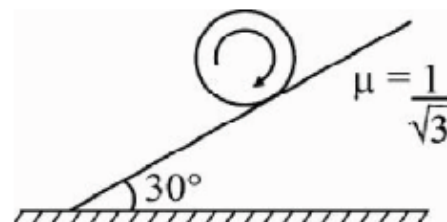


(A)  $2000R \left[ \frac{1}{2} + \ln \frac{4}{3} \right]$   
(B)  $500R (3 + \ln 4)$   
(C)  $500R \left( 2 + \ln \frac{16}{9} \right)$   
(D)  $100R \left( 1 + \ln \frac{16}{9} \right)$

11. The trajectory of a projectile in a vertical plane is  $y = ax - bx^2$ , where  $a$  &  $b$  are constants and  $x$  and  $y$  are respectively horizontal and vertical distance of the projectile from the point of projection. The maximum height attained by the particle and the angle of projection from the horizontal are respectively

(A)  $\frac{a^2}{b}, \tan^{-1}(2a)$   
(B)  $\frac{a^2}{4b}, \tan^{-1}(b)$   
(C)  $\frac{b^2}{2a}, \tan^{-1}(b)$   
(D)  $\frac{a^2}{4b}, \tan^{-1}(a)$

12. A disc is rotated about its axis with a certain angular velocity and lowered gently on a rough inclined plane as shown in the figure, then:



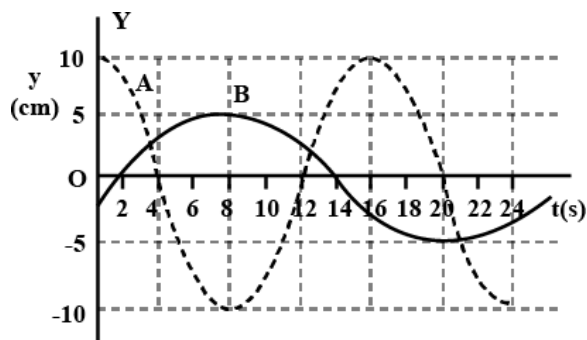
- (A) It will rotate at the position where it was placed and then will move downward  
(B) It will go downward just after it is lowered  
(C) It will go downward first and then climb up  
(D) It will climb upwards and then move downwards

13. The amplitude of a damped oscillator becomes one-fourth in one minute. The amplitude after 3 minute will be  $\frac{1}{x}$  times of the original.

Determine value of  $x$  :-

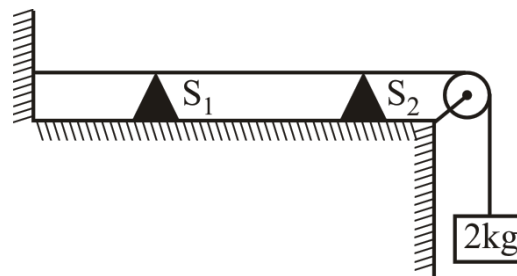
- (A) 12  
(B) 256  
(C)  $4^3$   
(D)  $\frac{1}{64}$

14. The following figure shows the displacement versus time graph for two particles A and B executing simple harmonic motions. The ratio of their maximum velocities is



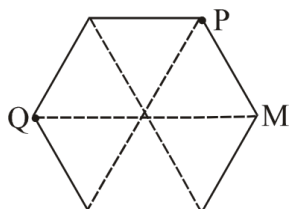
- (A) 3 : 1  
(B) 1 : 3  
(C) 1 : 9  
(D) 9 : 1

15. A sonometer wire vibrates with tuning fork of frequency 400 Hz. When the support  $S_2$  of sonometer wire is moved by a distance of 25cm, it again resonates with the given tuning fork. If mass of 2 kg was suspended by the wire find mass per unit length of wire : (There is no resonance in between)

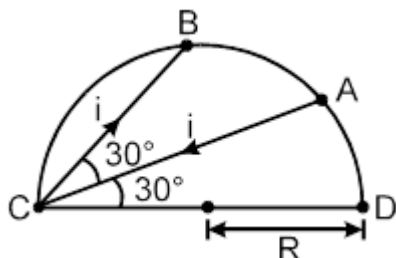


- (A) 0.5 gm/m  
(B) 1 gm/m  
(C) 1.5 gm/m  
(D) 2 gm/m
16. If speed (V), acceleration (A) and force (F) are considered as fundamental units, the dimension of mass will be :-
- (A)  $[V^0 A^0 F^1]$   
(B)  $[V^0 A^1 F^0]$   
(C)  $[V^0 A^{-1} F^1]$   
(D)  $[V^0 A^1 F^{-1}]$

17. Two sound source emitting sound of wavelength 1 m are located at points P and Q as shown in figure. All sides of the polygon are equal and of length 1m. The intensity of sound at M due to both the individual sources is  $I_0$ . What will be the intensity of sound at point M when both the sources are on.

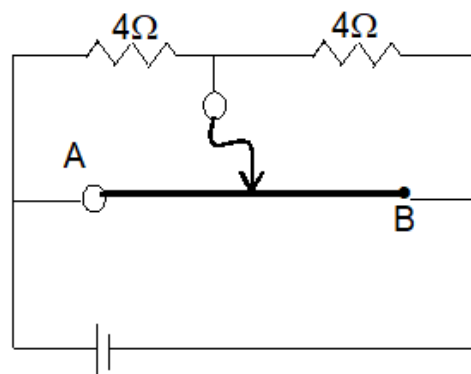


- (A)  $4I_0$  (B)  $2I_0$   
(C)  $I_0$  (D)  $(1/2)I_0$
18. A current carrying wire is placed in the grooves of an insulating semi circular disc of radius 'R', as shown. The current enters at point A and leaves from point B. Determine the magnetic field at point D.

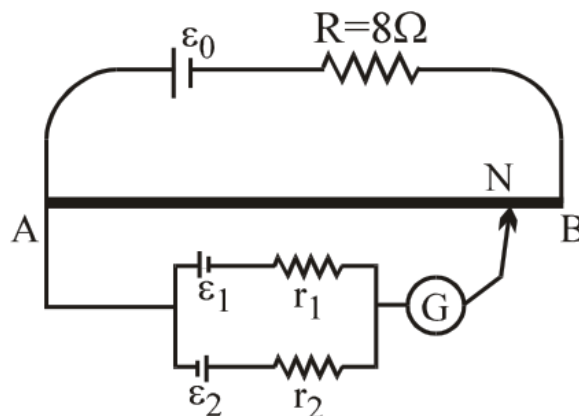


- (A)  $\frac{\mu_0 I}{8\pi R\sqrt{3}}$   
(B)  $\frac{\mu_0 I}{4\pi R\sqrt{3}}$   
(C)  $\frac{\sqrt{3}\mu_0 I}{4\pi R}$   
(D) none of these

19. The wire AB of a meter bridge continuously changes from radius  $r$  to  $2r$  from left end to right end. Where should the free end of galvanometer be connected on AB so that the deflection in the galvanometer is zero?



- (A) 50 cm (B) 66.66 cm  
(C) 40 cm (D) 33.33 cm
20. A battery of emf  $\epsilon_0 = 12V$  is connected across a 4m long uniform wire having resistance  $4\Omega/m$ . The cells of small emfs  $\epsilon_1 = 2V$  and  $\epsilon_2 = 4V$  having internal resistance  $2\Omega$  and  $6\Omega$  respectively, are connected as shown in the figure. If galvanometer shows no deflection at the point N, the distance of point N from the point A is equal to:



- (A)  $\frac{1}{6}m$  (B)  $\frac{1}{3}m$  (C) 25 cm (D) 50 cm

SECTION-II : (Maximum Marks: 20)

This section contains 10 questions. Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

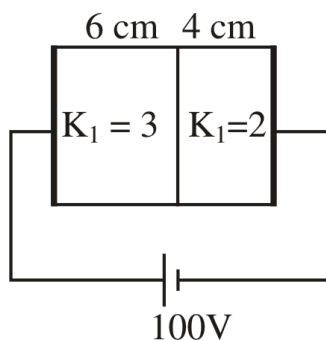
Answer to each question will be evaluated according to the following marking scheme:

Full Marks : +4 If correct answer is selected.

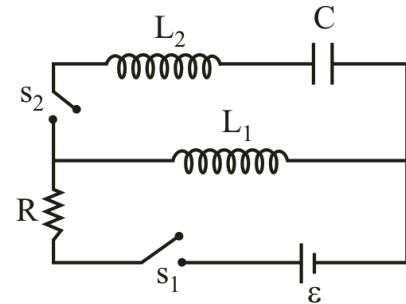
Zero Marks : 0 If none of the option is selected.

Negative Marks : -1 If wrong option is selected.

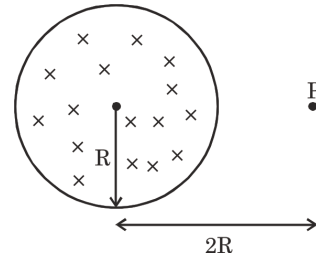
1. In a parallel plate capacitor the distance between the plates is 10cm. Two di-electric slabs of thickness 4cm and 6cm and di-electric constants  $K_1 = 3$  and  $K_2 = 2$  respectively are inserted and fill entire volume between the plates. A battery of EMF 100V is connected across the capacitor as shown in the figure. If the value of net bound surface charge density on the interface between dielectrics is  $\left(\frac{M}{N}\right) \epsilon_0 \times 10^2$  (S.I. units), find minimum value of  $(M + N)$  :



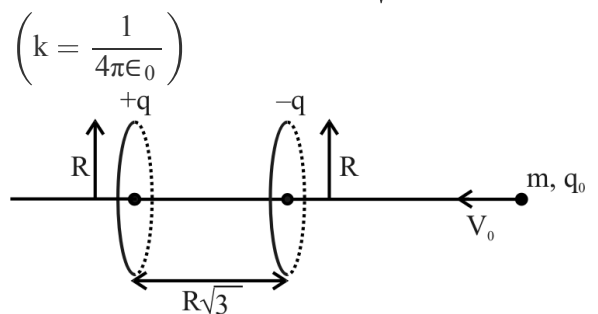
2. Diagram shows a circuit. Switch  $S_1$  was closed for a long time with  $S_2$  open. At  $t = 0$  switch  $S_2$  was closed &  $S_1$  was opened. The time period of L-C oscillation after  $t = 0$  is given by  $2\pi\alpha \times 10^{-4}$  s. Fill the value of  $\alpha$ . (Given  $C = 30\mu\text{F}$ ,  $L_1 = L_2 = 6\text{ mH}$ ).



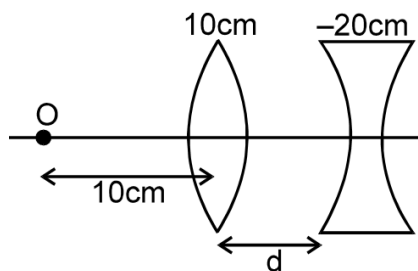
3. In the given diagram magnetic field changes with respect to time as  $B = 4t^2 - 2t$ ,  $R = 2\text{ m}$ . If a charge ( $m, q$ ) is placed at point 'P' at  $t = 2$  sec. Then acceleration of charge particle at  $t = 2\text{ sec}$ . is  $\frac{xq}{m}$  (S.I. Unit). Value of  $x$  will be.



4. Two uniformly charged co-axial, identical rings (Radius  $R$ ) are fixed as shown. A point charge  $q_0$  is projected with speed  $V_0$  along common axis of rings from large distance. Minimum value of  $V_0$  for which the charge particle can just pass through both rings is given as  $V_0 = \sqrt{\frac{kqq_0}{nmR}}$  find  $n$ .

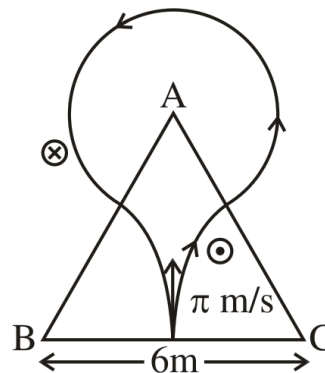


5. A simple microscope has a focal length of 5 cm. The magnification at the least distance of distinct vision is-
6. What should be the value (in cm) of distance  $d$  so that final image is formed on the object itself. (focal lengths of the lenses are as given in the figure).

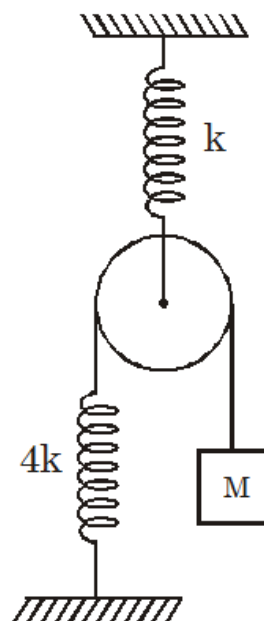


7. In a radioactive decay chain reaction,  ${}^{230}_{90}\text{Th}$  nucleus decays into  ${}^{214}_{84}\text{Po}$  nucleus. The ratio of the number of  $\alpha$  to number of  $\beta^-$  particles emitted in this process is \_\_\_\_\_.
8. In a hydrogen like atom an electron is orbiting in an orbit having quantum number  $n$ . Its frequency of revolution is found to be  $13.2 \times 10^{15}$  Hz. Energy required to free the electron from the atom from the above orbit is 54.4 eV. In time 7 nano - second the electrons jumps back to orbit having quantum number  $\frac{n}{2}$ .  $\tau$  be the average torque acted on the electron during the above process, then find  $\frac{\tau}{3} \times 10^{-27}$  in N - m.  
(Given  $\frac{h}{\pi} = 2.1 \times 10^{-34}$  J-s) Frequency of revolution of electron in the ground state of H - atom  $f_0 = 6.6 \times 10^{15}$  Hz and ionization energy of H - bond  $E_0 = 13.6$  eV

9. A positive charge 0.1 C and mass 0.1 kg is projected with velocity  $\pi$  m/s from mid point of line BC towards vertex A of an equilateral triangle of side 6m. The magnitude of magnetic field is  $\frac{\pi}{3}$  T inside and outside the triangle as shown in figure. Then time (in sec) after which the charged particle crosses side BC for first time, is



10. For the shown system find the maximum displacement (in cm) of the block if the system is released from natural length condition. Take  $m = 3\text{kg}$ ,  $k = 5100$  N/m. (Pulley is massless)



SECTION-I : (Maximum Marks: 80)

This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

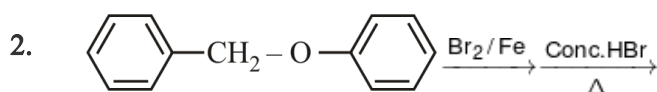
**Full Marks** : +4 If correct answer is selected.

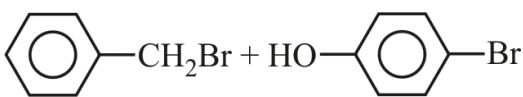
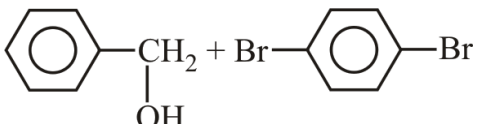
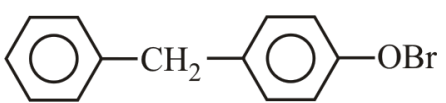
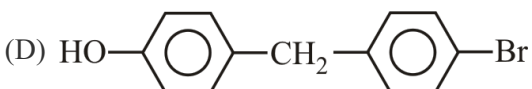
**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

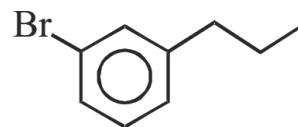
1. An organic compound A upon reacting with  $\text{NH}_3$  gives B. On heating, B gives C. C in presence of KOH reacts with  $\text{Br}_2$  to give D. D gives  $\text{CH}_3\text{CH}_2\text{NC}$  with KOH +  $\text{CHCl}_3$ . The structure of A is :

- (A)  $\text{CH}_3\text{COOH}$   
 (B)  $\text{CH}_3\text{CH}_2\text{CH}_2\text{COOH}$   
 (C)  $\text{CH}_3-\underset{\text{CH}_3}{\text{CH}}-\text{COOH}$   
 (D)  $\text{CH}_3\text{CH}_2\text{COOH}$

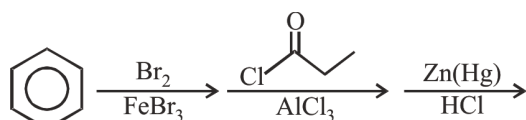
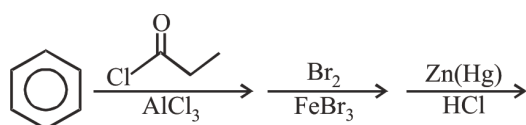
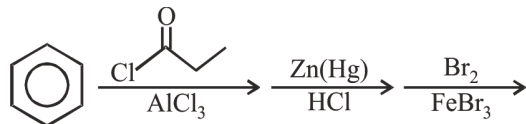
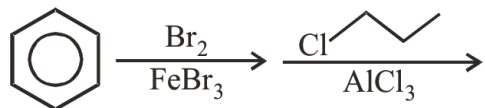


- (A)   
 (B)   
 (C)   
 (D) 

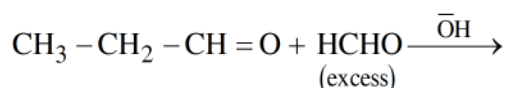
3. Which of the following reaction sequence would be the best for synthesizing the compound, 1-bromo-3-propylbenzene ?



1-Bromo-3-propylbenzene

- (A)   
 (B)   
 (C)   
 (D) 

4. Which of the following product mixture will contain major part ?



- (A)  $\text{CH}_3 - \underset{\text{CH}_2 - \text{OH}}{\text{CH}} - \text{CH} = \text{O}$
- (B)  $\text{CH}_3 - \underset{\text{CH}_2 - \text{OH}}{\overset{\text{CH}_2 - \text{OH}}{\text{C}}} - \text{CHO} + \text{HCOO}^-$
- (C)  $\text{CH}_3 - \underset{\text{CH}_2 - \text{OH}}{\overset{\text{CH}_2 - \text{OH}}{\text{C}}} - \text{CH}_2 - \text{OH} + \text{HCOO}^-$
- (D)  $\text{CH}_3 - \underset{\text{CH}_2 - \text{OH}}{\overset{\text{CH}_2 - \text{OH}}{\text{C}}} - \text{COO}^- + \text{CH}_3\text{OH}$

5. Aldehyde is not obtained as a product in which of the following reactions.

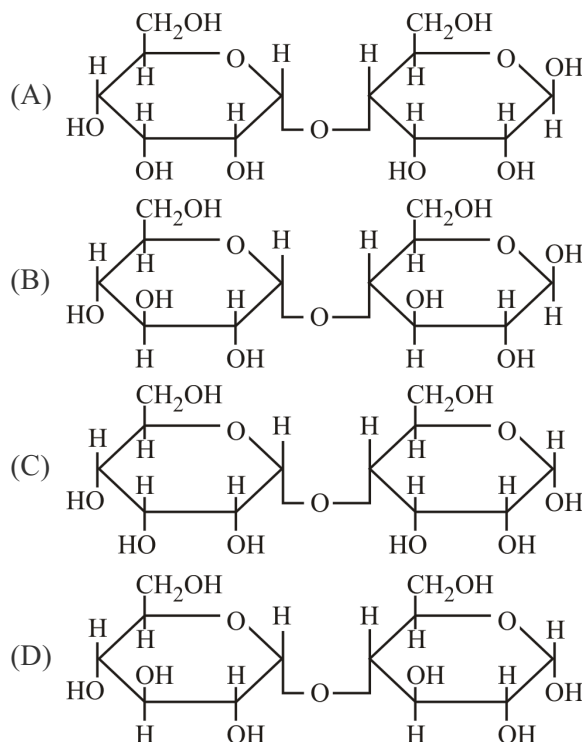
- (A)  $\text{H} - \overset{\text{O}}{\underset{\text{CH}_3}{\text{C}}} - \text{CH} - \text{COOH} \xrightarrow{\Delta}$
- (B)  $\text{CH}_3 - \text{C} \equiv \text{CH} \xrightarrow[\text{(ii) H}_2\text{O}_2 / \text{OH}^\ominus]{\text{(i) BH}_3 / \text{THF}}$
- (C)  $\text{CH}_3 - \overset{\text{O}}{\underset{\text{O}}{\text{C}}} - \text{Cl} \xrightarrow[\text{BaSO}_4]{\text{H}_2 / \text{Pd}}$
- (D)  $\text{CH}_3 - \underset{\text{CH}_3}{\text{CH}} - \text{OH} \xrightarrow[300^\circ\text{C}]{\text{Cu}}$

6. Match the list -I with list - II

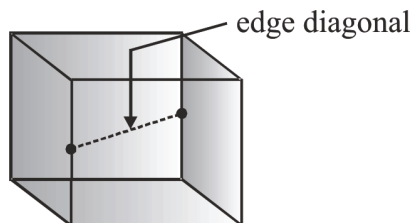
| List-I<br>(Class of Drug) |                      | List-II<br>(Example) |            |
|---------------------------|----------------------|----------------------|------------|
| (a)                       | Antacid              | (i)                  | Novestrol  |
| (b)                       | Artificial sweetener | (ii)                 | Cimetidine |
| (c)                       | Antifertility        | (iii)                | Valium     |
| (d)                       | Tranquilizers        | (iv)                 | Alitame    |

The correct code is :

- (A) (a) – (ii), (b) – (iv), (c) – (i), (d) – (iii)
- (B) (a) – (iv), (b) – (i), (c) – (ii), (d) – (iii)
- (C) (a) – (iv), (b) – (iii), (c) – (i), (d) – (ii)
- (D) (a) – (ii), (b) – (iv), (c) – (iii), (d) – (i)
7. Which of the following is correct structure of  $\alpha$ -anomer of maltose ?



8. Correct expression for determining packing fraction (P.F.) of an ideal NaCl-Type unit cell, if all ions along an edge diagonal are absent, as shown in the figure below. [Anions are at FCC lattice points,  $r_+$  = Cationic radius,  $r_-$  = Anionic radius]



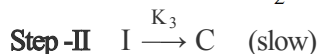
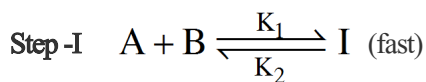
- (A)  $\frac{4}{3}\pi(r_+^3 + r_-^3)$   
 $16\sqrt{2}r_-^3$
- (B)  $\frac{4}{3}\pi\left(\frac{7}{2}r_+^3 + 4r_-^3\right)$   
 $16\sqrt{2}r_-^3$
- (C)  $\frac{4}{3}\pi\left(\frac{5}{2}r_+^3 + 4r_-^3\right)$   
 $16\sqrt{2}r_-^3$
- (D)  $\frac{4}{3}\pi\left(\frac{5}{2}r_+^3 + 4r_-^3\right)$   
 $16\sqrt{2}r_+^3$
9. For a weak monobasic acid, HA, the resistance of 0.4M - aq. HA solution is 100 ohm, at 25°C. The cell constant of the conductivity cell, in which resistance is measured, is  $0.8 \text{ m}^{-1}$ . If the molar conductance of HA at infinite dilution is  $0.4 \text{ ohm}^{-1} \text{ cm}^2 \text{ mol}^{-1}$ , the only correct information about HA solution is :
- (A) Conductance of solution is  $8 \times 10^{-3} \text{ ohm}^{-1} \text{ m}^{-1}$
- (B) Conductivity of solution is  $0.01 \text{ ohm}^{-1}$
- (C) The degree of dissociation of HA is 0.05
- (D)  $K_a$  of HA is 0.20

10. Assuming 100% ionization of electrolytes, the increasing order of the freezing point of the solution will be :-

- (A)  $0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2 < 0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4 < 0.10 \text{ mol kg}^{-1} \text{ KCl}$
- (B)  $0.10 \text{ mol kg}^{-1} \text{ KCl} < 0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4 < 0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2$
- (C)  $0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4 < 0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2 < 0.10 \text{ mol kg}^{-1} \text{ KCl}$
- (D)  $0.10 \text{ mol kg}^{-1} \text{ KCl} < 0.10 \text{ mol kg}^{-1} \text{ Ba}_3(\text{PO}_4)_2 < 0.10 \text{ mol kg}^{-1} \text{ Na}_2\text{SO}_4$
11. The R.M.S. speed of the molecules of a gas sample having density  $4 \text{ gm cm}^{-3}$  and pressure  $12 \times 10^7 \text{ N m}^{-2}$  is :
- (A)  $120 \text{ ms}^{-1}$
- (B)  $300 \text{ ms}^{-1}$
- (C)  $600 \text{ ms}^{-1}$
- (D)  $900 \text{ ms}^{-1}$
12. Calculate magnitude of work involved when 100 gm of Calcite form of  $\text{CaCO}_3$  is converted into it's Aragonite form of  $\text{CaCO}_3$  at 100 bar pressure. Given density of Calcite =  $2 \text{ gm/cc}$  and density of Aragonite =  $2.5 \text{ gm/cc}$ . [Atomic mass : Ca = 40]
- (A) 0.2 kJ
- (B) 0 kJ
- (C) 0.1 kJ
- (D) 0.5 kJ

13. For a complex reaction  $A + B \longrightarrow C$

Following mechanism is given :



If values of  $K_1$ ,  $K_2$  &  $K_3$  are  $2 \times 10^3 \frac{\text{litre}}{\text{mol}} \cdot \text{sec}^{-1}$ ,  $10^3 \text{ sec}^{-1}$  &  $0.01 \text{ sec}^{-1}$  respectively then value of overall rate constant for reaction ?

- (A)  $0.01 \frac{\text{mol}}{\text{litre}} \text{sec}^{-1}$  (B)  $0.01 \text{ sec}^{-1}$   
(C)  $0.02 \frac{\text{litre}}{\text{mol}} \cdot \text{sec}^{-1}$  (D)  $0.02 \frac{\text{mol}}{\text{litre}} \text{sec}^{-1}$

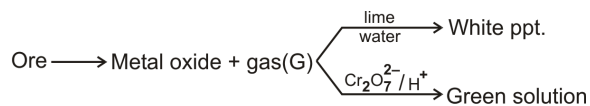
14. Which of the following is the correct electronic configuration of gadolinium ( $Z = 64$ )?

- (A)  $[\text{Xe}] 4f^7 5d^1 6s^2$   
(B)  $[\text{Xe}] 4f^6 5d^2 6s^2$   
(C)  $[\text{Xe}] 4f^8 6d^2$   
(D)  $[\text{Xe}] 4f^9 5s^1$

15. Pick up the correct statement.

- (A) CO which is major pollutant resulting from the combustion of fuels in automobiles plays a major role in photochemical smog  
(B) Classical smog has an oxidizing character while the photochemical smog is reducing in character  
(C) Photochemical smog occurs in day time whereas the classical smog occurs in early morning hours  
(D) During formation of smog the level of ozone in the atmosphere goes down

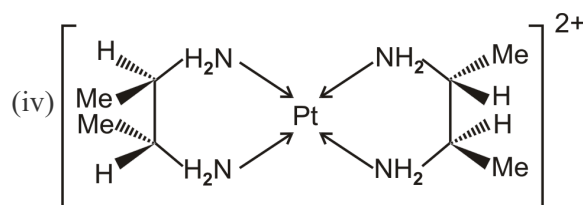
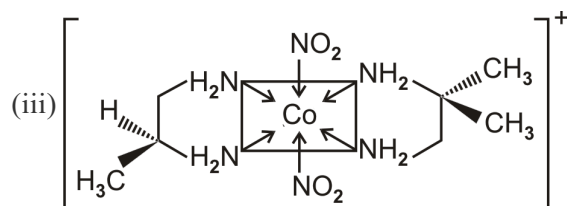
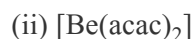
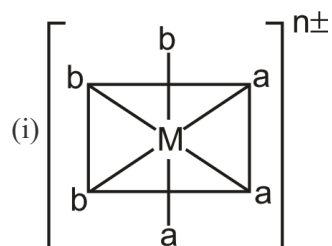
16. Consider the following steps of the extraction of Zn from its ore.



Which of the following option matches the above observation :

- (A) Ore is sphalerite  
(B) Ore is calamine  
(C) Ore is galena  
(D) Resulting gas (G) is  $\text{CO}_2$

17. Consider the following complexes.



In which of the following option only optical active complexes are present.

- (A) (i) (ii) (B) (i) (iii)  
(C) (ii) (iv) (D) (iii) (iv)

18. In which of the following molecule peroxy linkage(s) not present.

- (A)  $K_3CrO_8$
- (B)  $CrO_5$
- (C)  $H_2SO_5$
- (D)  $Na_3B_3O_6$

19. Choose the incorrect statement.

- (A)  $NH_3$  is more soluble compared to  $PH_3$  in water
- (B) Liquid  $I_2$  conducts electricity
- (C)  $F_2$  can be prepared by oxidising  $KF$  with  $KMnO_4$
- (D)  $XeO_3$  can be obtained by the hydrolysis of both  $XeF_4$  and  $XeF_6$

20. Assign true (T) of false (F) for following statements and select correct option for your answer –

- (I) I.P. of  $O_{(g)}$  is less than I.P. of  $O_{(g)}^-$
- (II) I.P. of  $Ne_{(g)}$  is greater than I.P. of  $Ne_{(g)}^+$
- (III) E.A. of  $O_{(g)}^+$  is greater than E.A. of  $O_{(g)}$
- (IV) I.P. of  $N_{(g)}$  is greater than I.P. of  $N_{(g)}^+$
- (A) F,F,T,F
- (B) T,T,T,T
- (C) T,T,T,F
- (D) F,T,F,T

### SECTION-II : (Maximum Marks: 20)

**This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.**

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

*Full Marks* : +4 If correct answer is selected.

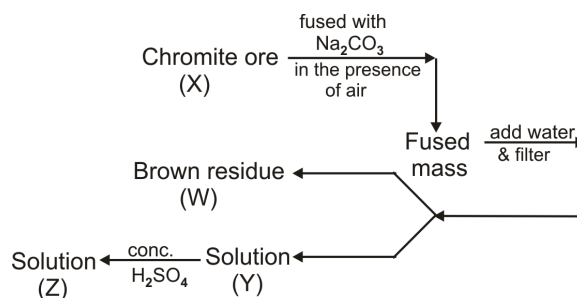
*Zero Marks* : 0 If none of the option is selected.

*Negative Marks* : -1 If wrong option is selected.

1. Count number of polymers which are condensation polymers as well as co-polymers :  
 (1) Polythene (2) Nylon-6, 6 (3) Nylon-6, 10  
 (4) Dacron (5) Bakelite  
 (6) Melamine formaldehyde resin  
 (7) Polyisoprene  
 (8) Styrene-butadiene rubber (Buna-S) (9) PAN  
 (10) Teflon
2. If H-atom of methyl cyclobutane is replaced by chlorine then how many mono chloroderivative will form (including stereo isomers)
3. When 0.15 g of an organic compound was analyzed using Carius method for estimation of bromine, 0.2397 g of  $AgBr$  was obtained. The percentage of bromine in the organic compound is \_\_\_\_\_. (Nearest integer)  
 [Atomic mass : Silver = 108, Bromine = 80]

4. 100 ml of 1M  $\text{AgNO}_3$  solution is electrolysed using Pt electrode by passing a current of 9.65A for 100 sec. Calculate pH of solution after electrolysis. [Neglect hydrolysis of  $\text{Ag}^+$  ions]
5. For the just coagulation of 250ml of  $\text{Fe}(\text{OH})_3$  sol, 2ml of 1M  $\text{Na}_2\text{SO}_4$  electrolyte is required. What is the coagulating value of  $\text{Na}_2\text{SO}_4$  electrolyte (in milli-moles per litre) ?
6.  $K_a$  of HCN is  $5 \times 10^{-10}$  at  $25^\circ\text{C}$ . For maintaining a constant pH of 9, the volume (in mL) of 5M KCN solution required to be added to 10mL of 2M HCN solution is :
7. What mass (in gm) of  $\text{N}_2\text{H}_4$  can be oxidised to  $\text{N}_2$  by 24.25 gm of  $\text{K}_2\text{CrO}_4$ , which reduces to  $\text{Cr}(\text{OH})_4^-$  ?  
[Atomic mass : Cr = 52, K = 39]
8. If for molecule  $\text{H}_4\text{P}_4\text{O}_{12}$
- Basicity of acid = a
  - Number of tetrahedral units with respect to 'P' = b
  - Number of ' $\pi$ ' bonds in one molecule = c
  - Number of atoms(s) with no lone pair in one molecule = d
- express your answer as 'abcd'.  
(if a = 4, b = 5, c = 6, d = 0 ; ans is 4560)

9. Following steps are involved in manufacturing of potassium dichromate :



- Oxidation number of Cr in X, Y and Z are the 1<sup>st</sup>, 2<sup>nd</sup> and 3<sup>rd</sup> digit of your answer respectively and 4<sup>th</sup> digit is the oxidation state of metal in 'W'.
10. For  $[\text{Cr}(\text{H}_2\text{O})_6]^{2+}$  ion the mean pairing energy (P) is found to be equal to  $23500 \text{ cm}^{-1}$ . The magnitude of  $\Delta_0$  is  $13900 \text{ cm}^{-1}$
- If
- x = magnitude of crystal field stabilising energy in more stable state (in  $\text{cm}^{-1}$ )
- y = magnitude of crystal field stabilising energy in less stable state (in  $\text{cm}^{-1}$ )
- Find the value of  $\left(\frac{x}{y}\right) \times 1000$

SECTION-I : (Maximum Marks: 80)

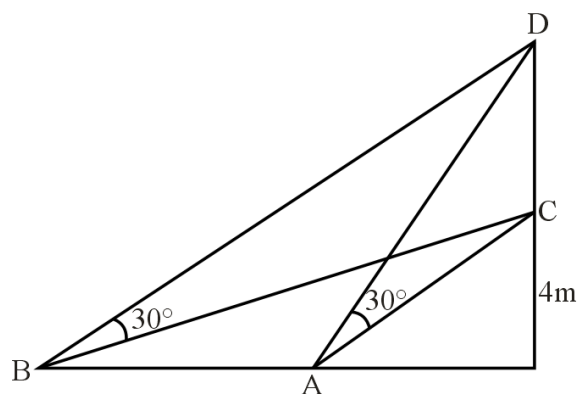
This section contains **20 questions**. Each question has 4 options for correct answer. Multiple-Choice Questions (MCQs) **Only one option is correct**. For each question, marks will be awarded as follows:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. Length of the flagstaff, on the top of a vertical pole 4m high, subtends equal angles  $30^\circ$  each at two points A, B on a straight line on the ground through the foot of the pole and on same side of pole. If distance of the point nearer to the foot of the pole is 4m, then length of the flagstaff is



- (A)  $6(\sqrt{3} - 1)$  m  
 (B)  $6(\sqrt{3} + 1)$  m  
 (C)  $4(\sqrt{3} + 1)$  m  
 (D) None of these
2. Angle between the curves  $y = \sqrt{|x| - 1}$  and  $y = \sqrt{5 - x^2}$  at their points of intersection is -  
 (A)  $\pi/6$  (B)  $\pi/3$   
 (C)  $\pi/2$  (D) None of these

3. The number of solutions of the equation  $\sqrt{2} \cos 3x - \cos 2x = \sqrt{3} \sin 2x + \sqrt{2} \sin 3x$  in  $\left[-\frac{4\pi}{3}, \frac{2\pi}{3}\right]$  is  
 (A) 3  
 (B) 4  
 (C) 5  
 (D) 6
4. Solution set of  $\cot^{-1} x > \sin^{-1} x$ , is  
 (A)  $-\infty < x < \sqrt{\frac{\sqrt{5}-2}{2}}$   
 (B)  $-1 \leq x < \sqrt{\frac{\sqrt{5}-1}{2}}$   
 (C)  $-1 \leq x < \sqrt{\frac{\sqrt{5}-2}{2}}$   
 (D) None of these
5. The value of  $\frac{\left(100 \int_0^{\pi/2} \theta \sin^{10} \theta d\theta\right) - 1}{\int_0^{\pi/2} \theta \sin^8 \theta d\theta}$  is  
 (A) 10  
 (B) 50  
 (C) 90  
 (D) 100

6. If  $y = \left(x - \sqrt{1+x^2}\right)^9$ , then  $(1+x^2)$

$$\frac{d^2y}{dx^2} + x \frac{dy}{dx} \text{ is}$$

(A)  $81y$

(B)  $-81y$

(C)  $162xy$

(D)  $-162xy$

7. Consider the statement p: There exists prime numbers that are not odd. Negation of statement 'p' is

(A) There exist prime numbers that are odd

(B) Not all prime numbers are even

(C) All prime numbers are odd

(D) None of these

8. A spherical iron-ball 20 cm in radius is coated with a layer of ice of uniform thickness that melts at a rate of  $50 \text{ cm}^3/\text{min}$ . When the thickness of ice is 5 cm, then the rate at which the thickness of ice decreases, is

(A)  $\frac{1}{50\pi} \text{ cm/min}$

(B)  $\frac{1}{2\pi} \text{ cm/min}$

(C)  $\frac{1}{18\pi} \text{ cm/min}$

(D)  $\frac{1}{72\pi} \text{ cm/min}$

9. If  $f(x) = 4x + \pi \sin x$ ,  $g(x) = f^{-1}(x)$  and  $g''(3\pi) = \pi/n$ , then the value of 'n' is

(A) 4

(B) 16

(C) 32

(D) 64

10. Let  $f(x) = x^4 + ax^3 + bx^2 + cx + d$ . If  $ac - 16d = 0$ ,  $g(x) = 12x^5 + 15ax^4 + 20bx^3 + 30cx^2 + 60dx + k$ ,  $a, b, c, d, k \in \mathbb{R}$  has 5 positive real zeroes and  $f(2) = 1$ , then the value of the sum of the zeroes of  $f(x)$  if this sum is greater than 4 is

(A) 12

(B) 15

(C) 20

(D) 30

11. The complete set of points where

$$f(x) = \frac{\sin x}{1 + |x| + x^2} \text{ is differentiable is}$$

(A)  $(-\infty, 0) \cup (0, \infty)$

(B)  $(-\infty, -1) \cup (1, \infty)$

(C)  $(-\infty, -1) \cup (-1, \infty)$

(D)  $(-\infty, \infty)$

12. The differential equation of the family of circles passing through the origin and whose centre lies on the line  $x = 2$  is
- (A)  $(x^2 - y^2 + 4x) \frac{dy}{dx} = 2y(x - 2)$
- (B)  $(x^2 + y^2 + 4x) \frac{dy}{dx} = 2y(x - 2)$
- (C)  $(x^2 - y^2 - 4x) \frac{dy}{dx} = 2y(x - 2)$
- (D)  $(x^2 + y^2 - 4x) \frac{dy}{dx} = 2y(x - 2)$
13. The probability of a shooter hitting a target is  $\frac{2}{3}$ . If he fires at a target 6 times, then the probability that he hits the target the second time in the fourth attempt is
- (A)  $\frac{8}{27}$
- (B)  $\frac{16}{81}$
- (C)  $\frac{4}{27}$
- (D) None of these
14. The value of  $\sqrt{2} \int \frac{\cos x dx}{\sin(x - \frac{\pi}{4})}$  is  
(Where  $c$  is arbitrary constant)
- (A)  $\log_e \left| \cos\left(x - \frac{\pi}{4}\right) \right| - x + c$
- (B)  $\log_e \left| \cos\left(x - \frac{\pi}{4}\right) \right| + x + c$
- (C)  $\log_e \left| \sin\left(x - \frac{\pi}{4}\right) \right| - x + c$
- (D)  $\log_e \left| \sin\left(x - \frac{\pi}{4}\right) \right| + x + c$
15. In the  $\triangle ABC$ , medians  $AD$  and  $BE$  are drawn. If  $AD = 4$ ,  $\angle BAD = \frac{\pi}{6}$  and  $\angle ABE = \frac{\pi}{3}$ , then area of the  $\triangle ABC$  is
- (A)  $\frac{64}{3}$
- (B)  $\frac{8}{3}$
- (C)  $\frac{16}{3}$
- (D)  $\frac{32}{3}\sqrt{3}$
16. Let  $a, b, c \in \mathbb{R}$  such that  $a^2 + b^2 + c^2 = 2bc$ ,  $a^2 + b^2 + c^2 \neq 0$ . The point among the following which lie on the line  $ax + by + c = 0$ , is
- (A)  $(-1, 2)$
- (B)  $(1, 2)$
- (C)  $(-1, -2)$
- (D)  $(2, -1)$
17. Area bounded by  $-8z^2 - 8\bar{z}^2 + 20z\bar{z} = 144$  is  
(where  $z$  is a complex number)-
- (A)  $\pi$
- (B)  $6\pi$
- (C)  $12\pi$
- (D)  $4\pi$
18. Mean and variance of a binomial probability distribution are 4 and  $\frac{4}{3}$  respectively, then the probability of atleast two success is equal to
- (A)  $\frac{552}{729}$
- (B)  $\frac{201}{243}$
- (C)  $\frac{298}{343}$
- (D)  $\frac{716}{729}$

19. The complete set of values of  $m$  so that there is atleast one value of  $x$  which satisfy the inequality  $25^x + 5^x < m$  is

- (A)  $m \in \mathbb{R}$   
 (B)  $m \in \left(-\infty, -\frac{1}{4}\right)$   
 (C)  $m \in (0, \infty)$   
 (D)  $m \in \left(-\infty, \frac{1}{4}\right)$

20. If the equation  $x^3 + ax^2 + bx + 216 = 0$  has three real roots in geometric progression, then

$\frac{a^2 + b^2}{ab}$  is equal to -

- (A)  $-\frac{37}{6}$   
 (B)  $-6$   
 (C)  $6$   
 (D)  $\frac{37}{6}$

**SECTION-II : (Maximum Marks: 20)**

**This section contains 10 questions Candidates have to attempt any 5 questions out of 10. If more than 5 questions are attempted, then only first 5 attempted questions will be evaluated.**

The answer to each question is an Integer Value Type questions.

For each question, enter the correct integer value (In case of non-integer value, the answer should be rounded off to the nearest Integer).

Answer to each question will be evaluated according to the following marking scheme:

**Full Marks** : +4 If correct answer is selected.

**Zero Marks** : 0 If none of the option is selected.

**Negative Marks** : -1 If wrong option is selected.

1. For what value of  $\lambda$ , the system of equations  $x + y + z = 6$ ,  $x + 2y + 3z = 10$ ,  $2x + 4y + \lambda z = 24$  is inconsistent :

2. Let  $A$  and  $B$  be two square matrices of same order such that  $A^3 + B^3 = 0$  and  $A^2B + AB^2 = 0$  but  $|A| \neq |B|$ . Then  $|A^2 - B^2|$  is equal to

3. The value of  $\lim_{x \rightarrow 0} \left( \frac{\int_0^{x^3} \sec^2 t^2 dt}{x \sin^2 x} \right)$  is

4. Suppose  $A_1, A_2, \dots, A_n$  are  $n$  sets each having 5 elements and  $B_1, B_2, \dots, B_{30}$  are thirty sets each with 3 elements, let  $\bigcup_{i=1}^n A_i = \bigcup_{j=1}^{30} B_j = S$  and each element of  $S$  belongs to exactly 10 of the  $A_i$ 's and exactly 9 of the  $B_j$ 's. Then  $n$  is equal to

5. Let  $A = \begin{pmatrix} 1 & -1 & 1 \\ 2 & 1 & -3 \\ 1 & 1 & 1 \end{pmatrix}$  and

$10B = \begin{pmatrix} \alpha & 2 & 2 \\ -5 & 0 & \beta \\ 1 & \gamma & 3 \end{pmatrix}$ . If  $B$  is the inverse of matrix  $A$ , then  $\alpha$  is -

6. The sum of all odd divisors (odd natural numbers) of 2520 is
7. If  $z^2 + z + 1 = 0$ , where  $z$  is complex number then the value of  $\left(z + \frac{1}{z}\right)^3 + \left(z^2 + \frac{1}{z^2}\right)^3 + \left(z^3 + \frac{1}{z^3}\right)^3 + \dots + \left(z^7 + \frac{1}{z^7}\right)^3$
8. If the area enclosed between the curve  $y = \log_e(2x + e)$  and the coordinate axes is  $K$ , then  $\frac{1}{K}$  is
9. If  $\lambda \int_{-\pi/4}^{\pi/4} \frac{e^{\cos x}}{1 + e^{\sin x}} dx = \int_{-\pi/4}^{\pi/4} e^{\cos x} dx$ , then  $\lambda$  equals -
10. Let two circles are passing through a fixed point  $(x_1, y_1)$  where  $x_1 > 0, y_1 > 0$  and both circle touch coordinate axes. If sum of their radii is equal to product of their radii, then  $(x_1 + y_1) - \frac{2}{\left(\frac{1}{x_1} + \frac{1}{y_1}\right)}$  is equal to

## Space for Rough Work



